

WiFi Fingerprinting-based Indoor Positioning with Machine Learning Algorithms

Luong Nguyen Thi
Faculty of Information Technology
Dalat University
Dalat, Vietnam
luongnt@dlu.edu.vn

Huy Quang Pham
Faculty of Mathematics and Informatics
Dalat University
Dalat, Vietnam
huyqp@dlu.edu.vn

Ninh Duong-Bao
College of Computer Science and Electronic Engineering
Hunan University
Changsha, China
duongbaoninh@hnu.edu.cn

Khanh Nguyen-Huu
Department of Electronics and Telecommunications
Dalat University
Dalat, Vietnam
khanhnh@dlu.edu.vn

Abstract—With the rapid advances of mobile devices, location-based services have received significant attention. Among the available services, finding the exact position of a person, especially indoors, is a challenging problem. For indoor environments, using WiFi-based technology for positioning purposes is reasonable due to its utilization of existing WiFi infrastructure. In this paper, we implement and compare the positioning results of three machine learning algorithms such as support vector machine, decision tree, and random forest. The algorithms are applied to a multi-condition WiFi fingerprinting dataset which was conducted in an office room where different environmental conditions are considered. The results show that the random forest achieves the best classification result with an accuracy of over 85%, while the two others get an approximate accuracy of 80%.

Keywords—WiFi fingerprinting, indoor positioning, machine learning, support vector machine, decision tree, random forest

I. INTRODUCTION

Nowadays, the Global Positioning System (GPS) has become a reliable and indispensable service to localize a person using a mobile device in outdoor environments. However, it is not true in indoor areas such as buildings since the satellite signals are blocked by walls or ceilings, thus, these signals are very weak indoors and cannot guarantee the same positioning accuracy as outdoors. For that reason, there requires the development of indoor positioning systems (IPS) to track the user's position indoors.

Currently, many technologies can be used for indoor positioning such as radio frequency identification (RFID) [1], Bluetooth [2], visible light communication (VLC) [3], vision [4], inertial sensors [5], etc. Due to the widespread of WiFi Access Points (APs) in indoor environments, there exist a lot of WiFi-based positioning systems that use the Received Signal Strength (RSS) values collected from the deployed APs to determine the user's position. The major challenge of these WiFi-based systems is the instability of the RSS values due to the effects of shadowing, multipath, or even the changes in surrounding environments such as the room temperature, the number of electrical devices, the number of working people, etc.

WiFi fingerprinting is one of the most popular and promising techniques for indoor positioning. This technique basically contains two phases: an offline training phase and an

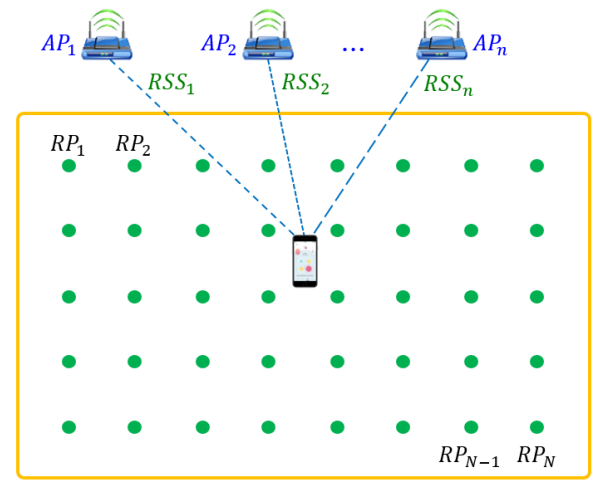


Fig. 1. RSS values collection.

online positioning phase. In the former phase, the RSS values are collected from available APs at different predefined reference points (RPs) in a setup area to make the fingerprints (i.e. sets of RSS values) for every RPs as shown in Fig 1. The fingerprint and the location of each RP together create the fingerprinting database (radio map). In the latter phase, the measured RSS values collected at an unknown position are compared and matched with the fingerprint of each RP in the database to find out the closest match, then the user's position is determined. Besides its utilization of WiFi infrastructure, the WiFi fingerprinting technique has another advantage as it does not require the line-of-sight condition from the APs, thus, it can be applied in complex environments where exist many obstacles such as the walls, doors, furniture, etc.

Generally, the matching algorithms in the online phase of the WiFi fingerprinting technique can be classified into two approaches: deterministic and probabilistic. RADAR [6] and Horus [7] were the very first systems that used the fingerprinting idea for indoor positioning. The first system used the K-nearest neighbors (KNN) which is one of the most popular algorithms of the deterministic approach. Meanwhile, the second system was based on the probabilistic approach which analyzed the statistical characteristics as well as the distribution of RSS values. More recently, following the deterministic approach, Ninh et al. [8] proposed a random statistical algorithm that firstly standardized the radio map in the offline phase, then applied the Mahalanobis distance to get

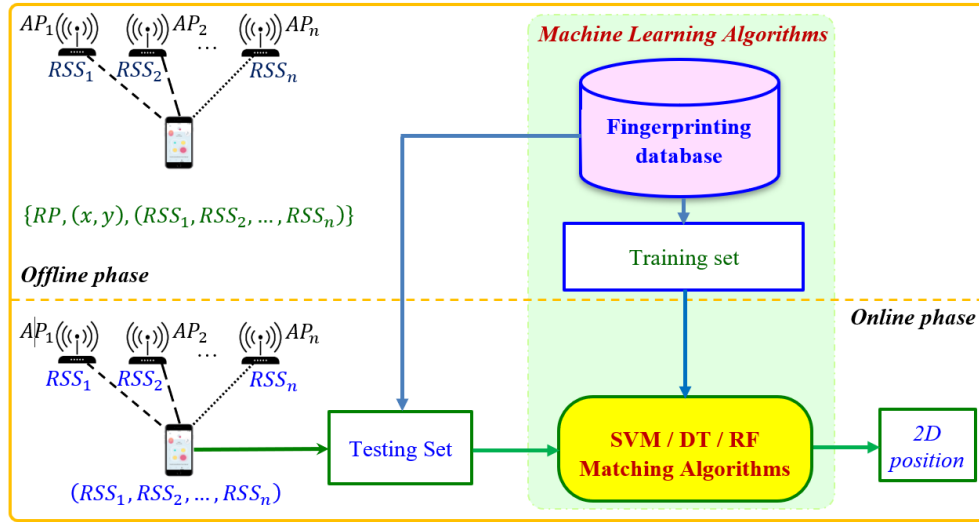


Fig. 2. System architecture.

the user's position instead of using the Euclidean distance which often works in the NN-based algorithms. Comparing the five different distance measures, Duong-Bao et al. [9] demonstrated that the basic Euclidean distance can be replaced by other distance measures to increase the positioning accuracy. The results revealed that the Chi-Squared distance was the best measure. Even when the authors changed the RSS collection settings in the offline phase such as changing the distance between two adjacent RPs or changing the number of available APs, the Chi-squared distance still kept its best results compared to other measures. Currently, the probabilistic approach also receives attention with different methods applied to solve the indoor positioning challenge. Kalman filter [10], particle filter [11], and hidden Markov models [12] are some famous algorithms used in this approach. To increase the positioning accuracy, Zhuang et al. [13] combined the tracking information from the inertial sensors as well as the WiFi fingerprinting using two Kalman filters. Moreover, with the same idea of combining different positional information from different algorithms such as WiFi fingerprinting, pedestrian dead reckoning, and some points of interest in indoor environments using an extended Kalman filter, Deng et al. [14] reduced the positioning error to under 1.5 m, which was a very promising result.

Over the past few years, machine learning algorithms have gained popularity in different aspects of our daily modern life, thus, these algorithms are also applied to indoor positioning to improve the positioning accuracy and enhance the robustness of the IPSs. To deal with the variation of the RSS values, which directly affects the performance of the WiFi fingerprinting, Rezgoui et al. [15] introduced a room-level positioning algorithm based on the support vector machine (SVM). From the experimental result, it was shown that the proposed algorithms achieved an accuracy of 98.75%. Bozkurt et al. [16] implemented and compared seven different machine learning algorithms such as KNN, decision tree (DT), Naïve Bayes, Adaboost, etc. The authors figured out that among the algorithms, KNN was the best one for solving classification problems with an accuracy of 99.7% for building and 98.5% for floor classifications, respectively. In [17], Gomes et al. proposed a hybrid random forest (RF) model to handle the fluctuations of the RSS values. From the experiments with seven setup APs, the high accuracy of 98.3% was reached using the K-fold cross-validation of 3. Meanwhile, in [18], Salamah et al. compared the SVM, DT,

and RF positioning results and concluded that SVM using the linear kernel surpassed the others with a 2-meter positioning error.

In this paper, we implement and compare the performance of three machine learning algorithms like SVM, DT, and RF. To evaluate the performance of each algorithm, we applied them to a free-accessed database that considered different environmental conditions when they collected the RSS values such as the number of electrical devices, the number of people, the period in a day, and the user's orientation, etc. We aim to analyze the classification accuracies of the aforementioned algorithms in a complicated indoor environment.

The remainder of the paper is organized as follows: Section 2 gives the material and methods. The experimental results are analyzed and discussed in Section 3. Finally, Section 4 concludes the paper.

II. MATERIAL AND METHODS

A. System Overview

Fig. 2 presents the system architecture of the WiFi fingerprinting with the machine learning classifiers. The system consists of two phases: an offline training phase and an online prediction phase. During the offline phase, the sets of RSS values are collected at different pre-defined RPs to create the fingerprinting database. Then, the training set and testing set are divided from the established database with a ratio of 9:1. The RSS values collected from available APs are used as the input features with the label is one RP position. Then, the RSS values will be put into the classifier for training. In the online prediction phase, the testing set is classified by applying different matching algorithms (i.e. the machine learning algorithms) to find out the user's position as one candidate among the whole RPs' positions.

B. Classification Algorithms

The three classification algorithms used in this paper are all supervised learning algorithms and each one is introduced as follows.

- Support vector machine (SVM) is one of the efficient machine learning algorithms which is used to solve the classification problem. This algorithm is firstly developed for binary classification, then expanded to

TABLE I. DT AND RF COMPARISON.

	DT	RF
Ease of implementation	Yes	No
Number of trees	One	Many
Memory	Small	Large
Features considered for a split at each decision node	All features	Random subset of features
Bootstrapping	No	Yes
Split	Best split	Best split

cover the multiclass classification in pattern recognition applications. SVM divides the dataset into two classes by finding the best hyperplane (i.e. the plane with the maximal margin between two classes) that separates all data points of one class from the ones of the other. This algorithm can cover both linear and nonlinear classification. The advantages of SVM are fast convergence speed, easy construction, and many adaption methods. Moreover, the SVM classifier is considered to have better accuracy compared to other classification algorithms [19].

- Decision tree (DT) is a well-known machine learning algorithm that creates a tree-like structure. The structure of the DT includes internal nodes, leaf nodes, and branches. Each internal node shows an attribute and it is associated with a relevant test for data classification. Leaf nodes are the nodes that represent class labels. Branches represent each of the possible results of the applied tests. The most advantages of DT are its ease of understanding and implementation.
- Random forest (RF) is first introduced by Breiman [20]. It is a classification algorithm that works by using multiple decision trees. Each tree learns simple rules extracted from the data. The complexity will be proportional to the increasing (deeper) of the trees. This algorithm attempts to overcome the overfitting problem of the basic DT. RF classifies instances based on multiple classifier's decisions, hence, it is also called an ensemble learning classification. The method uses the bagging idea to reduce the variance without increasing the bias. The majority voting rule will be executed after each DT made its own decision. RF's advantages are fast training and matching speed, stability, high classification accuracy, and the ability to work with large datasets. Table I displays the comparison between DT and RF algorithms at some criteria such as the ease of implementation, memory, bootstrapping, etc. to show the simplicity of DT compared to RF.

C. Dataset

In this paper, we use the WiFi fingerprinting dataset proposed by Duong-Bao et al. [21]. The major distinction of this dataset is that the authors considered different environmental conditions such as the density of people, the density of electrical devices, the user direction, the period in a day, etc. during the RSS values collection in the offline phase. This makes the RSS values at one RP change a lot, but this is practical in real indoor environments where the conditions can change much in a day. The dataset was created by a subject holding a smartphone to collect the RSS values in an office room that covered an area of 9.0 x 6.5 m². In this area, five APs were installed and 205 RPs were set up with the distance

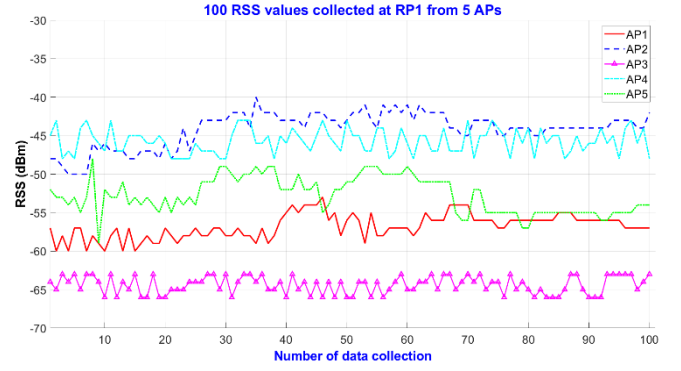


Fig. 3. Changes of RSS values over 100 scanning times at RP1.

between two adjacent RPs being 0.5 m. In the offline phase, the subject stood on each RP to collect the RSS values from the five APs 100 times over four months, thus, there were 20,500 sets of collected RSS values for 205 RPs used to create the fingerprinting database. Fig. 3 shows the changes of RSS values over 100 scanning times at one chosen RP. In the online phase, there were two test cases which were differed by environmental conditions, with the simpler setup for the first case and the more complicated setup for the second case. However, in this paper, we do not use the RSS values in the test cases but split the fingerprinting database into the training set and the testing set to evaluate the performance of the machine learning algorithms. The dataset's details can be found in [21]. All the implementations of the three classifiers and experimental analyses have been conducted under Python 3.8 with Numpy, Scipy, and Scikit-Learn libraries.

III. EXPERIMENTAL RESULTS

To evaluate the performance of the three aforementioned machine learning algorithms (i.e. SVM, DT, and RF) for the positioning purpose, we implement and apply them to the multi-condition WiFi fingerprinting dataset described in the above section. The fingerprinting database which was created in the offline phase will be divided into the training set and testing set with the ratio of 9:1, which means the K-fold cross-validation with K = 10 is applied. For instance, at each RP, the subject collected the RSS values 100 times, then we split these into 10 groups and each group will have an equal number of 10 observations. Then, we choose and shuffle nine groups for training and one group for testing. In the dataset, we have 205 RPs with 100 RSS scanning times for each RP, thus, there are a total of 20,500 sets of RSS values and they are divided into 18,450 sets for training and 2050 sets for testing.

Fig. 4 shows the mean accuracies from ten divided groups that are used for testing. From this figure, the RF algorithm generally achieves higher accuracies than others with the accuracies are all higher than 83.34 %, thus, it outperforms the classification results of other algorithms. The mean of mean accuracies of the three algorithms are illustrated in Fig. 5. As seen in this figure, the RF algorithm ranks in the first place with a mean accuracy of 87.13%, the runner-up belongs to the DT and the last one is the SVM with the mean accuracies staying approximately 80%. The reason for the superior performance of the RF may come from the randomly chosen RSS values from the radio map, which is suitable to handle the variations of the RSS values at one RP. The DT, however, uses a single tree so that it has a high variance in the classification results. The SVM performs terribly compared to both DT and RF because there exist many sets of RSS values (i.e. the fingerprints) that are similar to others but they belong

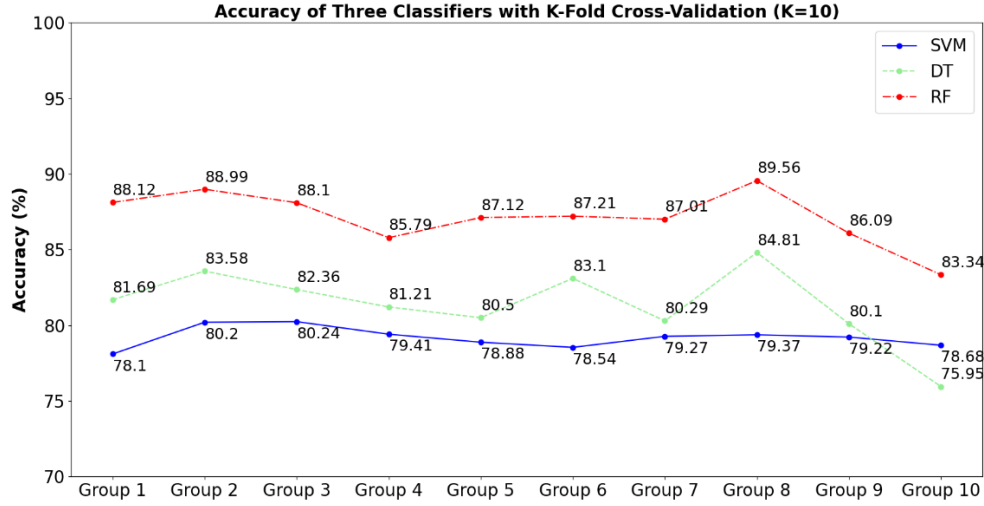


Fig. 4. Accuracy of the ten test groups from the 10-fold cross-validation.

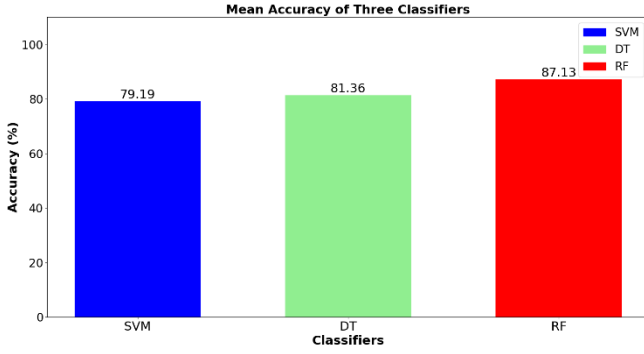


Fig. 5. Mean accuracies of three algorithms.

TABLE II. STATISTICAL COMPARISON OF THREE ALGORITHMS.

Algorithms	SVM	DT	RF
Max (%)	80.24	84.81	89.56
Min (%)	78.10	75.95	83.34
Mean (%)	79.19	81.36	87.13
Stdev (%)	0.65	2.32	1.69

to different RPs. This makes SVM unable to separate the RSS values to the right RP. Moreover, the high number of possible RPs (i.e. 205) also affects much to the performance of SVM since this algorithm is basically suitable to solve the classification problem with a minimum number of classes divided from the dataset.

Table II gives a statistical comparison of three algorithms. From this table, the RF algorithm is always the best one when applied to the multi-condition dataset due to its highest maximum classification accuracy (i.e. 89.56%) which is 5.03% and 10.41% higher than DT and SVM, respectively. Even the minimum accuracy of RF is just slightly lower than the maximum accuracy of DT and higher than the best result of SVM. This proves that the best classification algorithm belongs to RF. Meanwhile, the standard deviation of DT is the biggest one with 2.32% which confirms the high variance of classification accuracy of this algorithm compared to others.

IV. CONCLUSION

In this paper, we implement and analyze the performances of the three popular machine learning algorithms. These algorithms are tested with the multi-condition dataset which considered a bunch of environmental conditions while

collecting the RSS values in the offline phase. After running the experiments, the RF algorithm achieves the best classification result with the mean accuracy of 87.13%, which means this result is higher than the ones of DT and SVM 6.62% and 9.11%, respectively.

In the future work, we aim to test the performance of different machine learning algorithms in bigger areas such as in multi-floor buildings or in big shopping malls which have many rooms and floors. Moreover, we also want to implement and test the positioning potential of different deep learning algorithms such as convolutional neural networks or deep neural networks.

ACKNOWLEDGMENT

This work was supported in part by National Natural Science Foundation of China (NSFC) (61775054), and by National Natural Science Foundation of Hunan Province (grant no. 2020JJ4210).

REFERENCES

- [1] F. Seco and A. R. Jiménez, "Smartphone-Based Cooperative Indoor Localization with RFID Technology," *Sensors*, vol. 18, no. 1, 2018, pp. 266-289.
- [2] X. Li, J. Wang, and C. Liu, "A Bluetooth/PDR Integration Algorithm for an Indoor Positioning System," *Sensors*, vol. 15, no. 10, 2015, pp. 24862-24885.
- [3] M. Afzalan and F. Jazizadeh, "Indoor Positioning Based on Visible Light Communication: A Performance-based Survey of Real-world Prototypes," *ACM Computing Surveys*, 2019, pp. 1-6.
- [4] A. Xiao, R. Chen, D. Li, Y. Chen, and D. Wu, "An Indoor Positioning System Based on Static Objects in Large Indoor Scenes by Using Smartphone Cameras," *Sensors*, vol. 18, no. 7, 2018, pp. 2229-2246.
- [5] K. Nguyen-Huu and S.-W. Lee, "A Multi-Floor Indoor Pedestrian Localization Method Using Landmarks Detection for Different Holding Styles," *Mobile Information Systems*, vol. 2021, 2021, pp. 1-21.
- [6] P. Bahl and V. N. Padmanabhan, "RADAR: an in-building RF-based user location and tracking system," in *Proceedings IEEE INFOCOM 2000.*, vol. 2, 2000, pp. 775-784.
- [7] M. Youssef and A. Agrawala, "The Horus WLAN location determination system," in *Proceedings of the 3rd international conference on Mobile systems, applications, and services*, Seattle, Washington, 2005, pp. 205-218.
- [8] D. B. Ninh, J. He, V. T. Trung, and D. P. Huy, "An effective random statistical method for Indoor Positioning System using WiFi fingerprinting," *Future Generation Computer Systems*, vol. 109, 2020, pp. 238-248.

- [9] N. Duong-Bao, J. He, L. N. Thi, and K. Nguyen-Huu, "Analysis of Distance Measures for WiFi-based Indoor Positioning in Different Settings," in *2022 2nd International Conference on Innovative Research in Applied Science, Engineering and Technology (IRASET)*, 2022, pp. 1-7.
- [10] Z. Chen, H. Zou, H. Jiang, Q. Zhu, Y. C. Soh, and L. Xie, "Fusion of WiFi, smartphone sensors and landmarks using the Kalman filter for indoor localization," *Sensors*, vol. 15, no. 1, 2015, pp. 715-732.
- [11] X. Wang, G. Chen, M. Yang, and S. Jin, "A Multi-Mode PDR Perception and Positioning System Assisted by Map Matching and Particle Filtering," *International Journal of Geo-Information*, vol. 9, no. 2, 2020, pp. 93-116.
- [12] O. P. Babalola and V. Balyan, "WiFi Fingerprinting Indoor Localization Based on Dynamic Mode Decomposition Feature Selection with Hidden Markov Model," *Sensors*, vol. 21, no. 20, 2021, pp. 6778-6791.
- [13] Y. Zhuang, Y. Li, L. Qi, H. Lan, J. Yang, and N. El-Sheimy, "A Two-Filter Integration of MEMS Sensors and WiFi Fingerprinting for Indoor Positioning," *IEEE Sensors Journal*, vol. 16, no. 13, 2016, pp. 5125-5126.
- [14] Z.-A. Deng, G. Wang, D. Qin, Z. Na, Y. Cui, and J. Chen, "Continuous Indoor Positioning Fusing WiFi, Smartphone Sensors and Landmarks," *Sensors*, vol. 16, no. 9, 2016, pp. 1427-1447.
- [15] Y. Rezgoui, L. Pei, X. Chen, F. Wen, and C. Han, "An Efficient Normalized Rank Based SVM for Room Level Indoor WiFi Localization with Diverse Devices," *Mobile Information Systems*, vol. 2017, 2017, pp. 1-20.
- [16] S. Bozkurt, G. Elibol, S. Gunal, and U. Yayan, "A comparative study on machine learning algorithms for indoor positioning," in *2015 International Symposium on Innovations in Intelligent Systems and Applications (INISTA)*, 2015, pp. 1-8.
- [17] R. Gomes, M. Ahsan, and A. Denton, "Random Forest Classifier in SDN Framework for User-Based Indoor Localization," in *2018 IEEE International Conference on Electro/Information Technology (EIT)*, 2018, pp. 537-542.
- [18] A. H. Salamah, M. Tamazin, M. A. Sharkas, and M. Khedr, "An enhanced WiFi indoor localization system based on machine learning," in *2016 International Conference on Indoor Positioning and Indoor Navigation (IPIN)*, 2016, pp. 1-8.
- [19] C. J. C. Burges, "A Tutorial on Support Vector Machines for Pattern Recognition," *Data Mining and Knowledge Discovery*, vol. 2, no. 2, 1998, pp. 121-167.
- [20] L. Breiman, "Random Forests," *Machine Learning*, vol. 45, no. 1, 2001, pp. 5-32.
- [21] N. Duong-Bao, J. He, T. Vu-Thanh, L. N. Thi, L. Do Thi, and K. Nguyen-Huu, "A Multi-condition WiFi Fingerprinting Dataset for Indoor Positioning," in *Artificial Intelligence in Data and Big Data Processing, Cham*, 2022, pp. 601-613.